NIHR | National Institute for Health and Care Research

Summary

Reference Number	NIHR306483
Lead Applicant	Dr Thomas Rance
Research Title	Exploring the association between adventurous play and injury in childhood
Plain English Summary	**Background** Playing adventurously is the most common way that children exercise. Adventurous play (AP) includes activities like climbing up trees or cycling really fast. Not only do children enjoy playing like this but it is good for their health. Children who play adventurously are more active which is really important for a healthy childhood. AP can also teach children how to manage future risks and help them develop skills which are important for when they grow up. It also improves children's mental health and can help anxiety. Children now spend less time playing adventurously; this is bad for their health. I have shown in my previous work that this is because there are worries about them being hurt. However, there is no research that shows that children who play adventurously are more likely to be injured. In fact no recent research has been done to look at why children get hurt in the UK or how to stop it. This means we are stopping children from having a fun and healthy childhood for no good reason. **Aims** This research aims to find out whether children are more likely to hurt themselves during AP in comparison to other activities. It will also find out the reasons for children hurting themselves and what makes this more likely. It will also look at the numbers of children who have very bad injuries and work out what make this more likely. **Methods** I will do three different projects to answer the research question. 1. I will ask parents about the way their children play, the injuries their children had, how they happened, and whether children who play adventurously are more likely to be hurt. 2. I will gather information from children in Accident & Emergency departments about how they were hurt, what activities they were doing and what may have caused it. 3. I will use existing information that has been collected about serious and life changing injuries to see if there are common reasons and to see if AP is a major cause of these bad injuries. **Patient and Public Involvement and Engagement** I have talked to children, parents, doctors, nurses and charities to understand what they think about AP. Children don't seem to worry about hurting themselves, but would like information about how safe certain activities are. Parents are much more worried, about their children being hurt and about getting in trouble and blamed for their children's injuries. They have asked for me to also look at who is more likely to get hurt and specifically the effect of supervision, safety equipment like helmets and medical problems like autism and ADHD. Parents are also worried about really bad injuries which is the reason for the third project. The final details of the research will be written with parents and children to make sure the information is clear and relevant to their lives. I will work with a specific group at the University that specialise in this type of work called CA:RING. **Results** The project will bring together this knowledge to help children, parents, childcare workers, teachers and healthcare professionals have a healthy, fun and safe childhood. The results will be fed back to the people who have answered questions as part of the research, parenting groups, schools, nurseries, healthcare departments as well as other researchers at scientific conferences. It is my hope that we can learn how to safely encourage children to play as much as possible, improving children's health. Following this, I would like to use this work to look at ways to prioritise play and learn more about children's injuries.

Our purpose for collecting this data is to communicate with you about your application and have the necessary information to evaluate you for a grant. The data we collect here is collected in the public interest. Information provided here may be subject to Freedom of Information requests.

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The NIHR privacy policy includes further information including ways we may use your data, our contact details and details on your individual rights regarding how your data is used. Your data may be shared across the NIHR, including with other coordinating centres, to allow the application to be managed and for statistical analysis, and with external grant reviewers as part of the process for managing the allocation of a grant. This notice is under constant review and will be updated and / or revised based on that review as appropriate.

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1. Application Summary Information

Host organisation (which will administer any award)
University of Cambridge

Partner organisation (if applicable)	

Application Type

Strategic Theme	Public health

Partnership Fellowship	N/A

WTE % Please select which WTE option you would prefer
1.00

Duration Doctoral Fellowship is a 3 year award and based on your WTE preference the duration in months is calculated as follows
36

Research Title	Exploring the association between adventurous play and injury in childhood

Research Type	Epidemiological

Proposed start date if grant awarded	01/10/2025

2. Applicant CV

A CV for the Applicant is a mandatory component of this application; please ensure this is completed fully.
Please note that some of the CV questions are automatically completed from information you have provided in the 'Update CV' section of the 'Manage My Details' page. These can be updated via the ARAMIS Home Page or by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

The 'Publication Record' section of the form is automatically populated from publications added into the "My Research Outputs" page of your ARAMIS account.

Degrees and Professional Qualifications

From	To	Qualification	Subject	Country	Institution	Class	Department
02/2019	09/2021	MRCPCH	Membership to Royal College of Paediatrics	United Kingdom (England)	Royal College of Paediatrics	Pass	Paediatrics
09/2014	08/2015	BSc(Hons)	International Health	United Kingdom (England)	University of Leeds	2.1	School of Medicine
09/2011	07/2017	MBChB	Medicine	United Kingdom	University of Leeds	Pass	Medicine

Present and previous positions
(including from/to dates, job title and employer)

From	To	Position	Organisation	WTE % Research
09/2024	03/2025	Paediatric Registrar in Rheumatology (Part time academic role)	Great Ormond Street Hospital For Children NHS Foundation Trust	20
03/2024	09/2024	Paediatric Registrar	Guy's and St.Thomas' Hospitals Trust	0
09/2023	03/2024	Paediatric Registrar	Lewisham and Greenwich Hospitals Trust	0
09/2022	09/2023	Pre-doctoral research fellow in Paediatric Emergency Medicine	Kings College Hospital	25
03/2022	09/2022	Doctor - ST3 Paediatrics	Great Ormond Street Hospital	0
09/2021	03/2022	ST3 Paediatrics	Royal Brompton Hospital	0
03/2021	09/2021	ST2 Paediatrics	Imperial College	0
03/2020	03/2021	Research Fellow	Great Ormond Street	20

Research Grants held

Title of Grant	Role in the Research Grant	Total Grant £	Source of Grant (Organisation Name)	Duration (In Months)	Start Date of Funding	End Date of Funding
NHSE - Postgraduate Paediatric Training Evaluation	PI - Grant to facilitate the evaluation of paediatric training in London wide. Yearly administered grant to fund data collection and analysis. Additional grant awarded for longitudinal analysis from 2017-2022.	23000	NHSE	12	01/09/2023	02/09/2024

Title of Grant	Role in the Research Grant	Total Grant £	Source of Grant (Organisation Name)	Duration (In Months)	Start Date of Funding	End Date of Funding
Predoctoral Fellowship	Predoctoral fellowship awarded locally to facilitate paediatric emergency medicine research in preparation for this grant application.	18000	King's College Hospital London	12	05/09/2022	31/08/2023

Publication Record

(include the list of authors, article title, journal name, and volume)
This information can be edited via the "My Research Outputs" page of your ARAMIS account.

Publications
Rance, T.. (2024) 'The art of paediatrics: a history of play in hospitals', *Archives of disease in childhood*.
Ferin, K., Hufflett, T., Rance, T. and Round, J.. (2024) 'Making medical education great again: analysing trainee feedback to identify key drivers of trainee satisfaction and improve experience.', *Archives of disease in childhood. Education and practice edition*.
Rance, T., Ramchandani, P. and Hesketh, KR.. (2024) 'Risky play: our children need more.', *Archives of disease in childhood*.
Norrish, G., Rance, T., Montanes, E., Field, E., Brown, E., Bhole, V., Stuart, G., Uzun, O., McLeod, KA., Ilina, M., Adwani, S., Daubeney, P., Delle Donne, G., Linter, K., Jones, CB., Bharucha, T., Cervi, E. and Kaski, JP.. (2021) 'Friedreich's ataxia-associated childhood hypertrophic cardiomyopathy: a national cohort study.', *Archives of disease in childhood*, 107(5).
Rance, T. and Friend, AJ.. (2018) 'There is no role for routine annual echocardiography in patients with repaired dextro-transposition of the great arteries without other cardiac anomalies.', *Archives of disease in childhood. Education and practice edition*, 104(4).
'Rance T, Bucke D, Kiemle-Gabbay J, et al. Improving safety-netting advice in a paediatric A&E – utilising public and patient involvement methodologies to inform strategy for quality improvement. Archives of Disease in Childhood 2023;108:A28-A29.',
'Rance, T. (2023). 'Retrospective cohort analysis of play related injuries – a matter of priorities.' Archives of Disease in Childhood ;108:A2-A3.',
'De Vijver, MV., Rance, T., Perkins, A., Ferin, K., Callaghan, B. and Round, J..(2023) Excellence exchanges to improve regional paediatric training. Archives of Disease in Childhood, 108:A454-A455.',
'Ferin, K., Rance, T., Oram, R., (2024). 'Evaluation of trainees' supporting professional activity (SPA) time allocation within the London School of Paediatrics.' Archives of Disease in Childhood, 109, A421-A422.',
'Rance, T., Ferin, K., Evans, K., Hyrapetian, M., Longley, C., Oram, R., Perkins, A., Tran, S., van De Vijver, M. (2023). 'London School of Paediatrics Survey – Learning points from the 10th annual survey: March 2022-September 2022.' Archives of Disease in Childhood. doi: 10.1136/archdischild-2023-rcpch.515.',
'Ross, K., Rance, T., Round, J., et al. (2024). 'MMEGA: making medical education great again.' Archives of Disease in Childhood, 109, A412-A413.',
'Perkins, A., van de Vijver, M., Ferin, K., Jenkins, C., Soo, A., Rance, T., Stilwell, A., Hyrapetian, M., Petrides, C.,(2021). 'Paediatric training during the COVID-19 pandemic: the voice of 638 trainees.' Archives of Disease in Childhood. doi: 10.1136/archdischild-2021-rcpch.672.',
'Rance, T., van de Vijver, M., Le Geyt, J.L. (2021). Hospital at home: child and parental experiences of ambulatory antibiotics following review and discharge from accident & emergency. Archives of Disease in Childhood. DOI: 10.1136/archdischild-2021-rcpch.639.',

Relevant Prizes, Awards and other Academic Distinctions

Please note the information for additional prizes or awards is automatically populated from information provided in the "Update CV" section of the "Manage My Details" page, which you can update by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

Date (mm/yyyy)	Source	Prize	Role of Applicant
01/09/2024	BSPCH	William Cadogan Prize	Lead
01/03/2024	RCPCH	Michael Blacow Prize - Best abstract RCPCH conference	Project co-lead
01/07/2021	RCPCH	Top mark nationwide Clinical Membership Examination to the RCPCH.	Recipient

ORCID iD 0000-0002-0250-5358

3. Applicant Research Background

Applicant Professional Background	Medically Qualified

Specialty (if Medically Qualified)
Paediatrics

Do you have a GMC number?	Yes

GMC number Please supply a valid GMC number	7562595

Please describe your research career to date

Introduction

I am a paediatric registrar and academic clinician at Great Ormond Street Hospital. My research interests focus on investigating the barriers to physical activity which pairs with my clinical interests in paediatric public health and community paediatrics. My academic excellence and research has been recognised with prizes from the RCPCH for achievements in my postgraduate exams and for research presentations. It has additionally been recognised by the associated paediatric societies for research and essays. My next rotation is an academic role in paediatric public health which will further equip me with skills prior to entering this fellowship in October 2025.

Clinical and Research experience

My academic work is currently conducted at two sites. Firstly, at the MRC epidemiology unit at the University of Cambridge where I propose to conduct my PhD. Here, I have conducted research exploring barriers to an active childhood, specifically looking at the parent reported barriers to adventurous play (AP). I am additionally co-author to a systematic review exploring the tools to quantitatively measure AP. This has allowed me to further develop my research skills in study design and statistical analyses and gain valuable experience in the research environment. I have developed a productive relationship with my proposed supervisors publishing peer-reviewed papers together whilst developing this PhD proposal.

Secondly, I hold a clinical academic position at the NIHR Clinical Research Facility at the Institute of Child Health where I am co-PI for the SELECT-JIA and STAR-JIA trials. These studies are multi-centre, randomised comparative studies for treatment of Juvenile Idiopathic Arthritis. In this role I have gained valuable experience of study coordination, patient enrolment, data collection and engaging with relevant research protocols. Whilst this role is not directly relevant to my proposed research, it has equipped me with valuable skills and demonstrates my commitment to clinical academia.

Prior to this I have completed a pre-doctoral research fellowship King's College London, developing this research proposal and enhancing my research skills to the level expected of a PhD applicant. During this fellowship I piloted appropriate research methodologies, gained knowledge of statistical software (R & Stata), and interrogated relevant databases to this proposal. This fellowship also allowed me to co-develop this PhD proposal using PPIE to incorporate children's voices as well as their parents. This taught me valuable skills in co-production which I intend to use throughout my project. The above work has already been acknowledged through presentations at local and national conferences and publication in peer-reviewed journals.

I have undertaken a series of previous roles in paediatric research departments to build a foundation of research skills. This includes an academic clinical fellowship at the Institute of Inherited Cardiovascular Diseases at UCL and Great Ormond Street Hospital conducting research investigating the hypertrophic cardiomyopathy of Friedreich's ataxia. This was the largest research project of its kind, published in the BMJ and presented orally at

the British Cardiovascular Society. Here I gained experience in managing large data sets, writing research papers and skills in oral presentations at major national conferences.

Teaching experience and qualifications

During my training I have designed and delivered undergraduate and postgraduate medical education. This includes question writing for the MBChB programme at the University of Leeds and examining in practical examinations. I was formally recruited to the faculty of Medical Education at the University of Leeds delivering lectures to undergraduate students, as well as small group tutoring. For this I received a nomination for the excellence in education award. I additionally teach at the University of Cambridge having recently designed and delivered a student self-selected module about play which has direct correlations to this proposal.

My formal educational qualifications include instructor status for the European Paediatric Advanced Life Support and Neonatal life support courses for the Resus Council.

Leadership experience & previous grants

I consult on the executive board to Starlight charity, collaborating with the senior leadership team, NHSE executives and third-sector partnerships to improve hospital play facilities nationwide. We have recently published our recommendations, informed by research conducted by the PEDAL department at the University of Cambridge. This will directly inform the commissioning of play services in the NHS. I have also contributed to the national play commission. My essay on play in hospital in the UK received the 2024 William Cadogan prize.

I am a member of the senior executive committee of the London School of Paediatrics. This work involves innovative quality improvement projects, data collection and analysis, and collaborating with executive and clinical staff across the London deanery to improve postgraduate paediatric education. A 5-year longitudinal analysis from this project was recently published and this work was awarded the Michael Blacow prize at the RCPCH conference. This work was facilitated by a small grant of £3,000 per year for the past 5 years plus a £5,000 supplementary grant for the longitudinal analysis.

Suitability for Fellowship

My previous research experience, active engagement in the host department, my pre-doctoral fellowship and my established relationship with my proposed supervisors has provided me with the skills to undertake this research. This working relationship has already produced tangible outputs.

My experience highlights my potential and commitment to be a future health research leader, and my ability to balance responsibilities. Furthermore, my academic achievements including the highest mark in the country in the RCPCH Clinical Exam in 2021 and recent academic prizes (Michael Blacow Prize and William Cadogan prize) demonstrate that I have the aptitude to succeed in a rigorous PhD programme.

Impact of this fellowship
This PhD will serve as a catalyst in my academic career enabling me to expand my knowledge of epidemiology and paediatric public health. The fellowship offers opportunities to develop into an independent researcher with international research collaborations. These experiences will be facilitate the development of an academic clinical lectureship and pursuing an academic clinical career in subspecialty paediatric public health. I wish to pursue further research to improve physical activity in children to improve long term health outcomes.

Have you already registered for a PhD?	No

Has this application, or a similar application, previously been submitted to this or any other funding body?
Yes

If Yes, provide details

NIHR Doctoral Fellowship R10 – Unsuccessful

I'm grateful for the reviewer's previous comments and suggestions. Whilst reviewers found the topic interesting and worthwhile for research there was room for improvement in the proposal. I have taken the subsequent two years to address this feedback as summarised below.

Person

Comments	Improvements
Limited time in career for research	In the time since the previous submission, I have taken on an academic clinical role at Great Ormond Street in the NIHR clinical research facility. I also have a formal role at the MRC epidemiology unit at the University of Cambridge where I intend to conduct this PhD where I have established relationships with my supervisors. I developed an interest in research late in my clinical career and I acknowledge my previous lack of formal research training prior to my pre-doctoral fellowship. However, I believe this demonstrates my commitment, having undertaken the previously described training and proposal in my own time.
Limited research outputs	I have published several papers and given multiple oral presentations at national conferences in the time between these applications relevant to my subject area. I have also received two academic prizes for this research..

Project Proposal

Comments	Improvements
Methodologies required more clarity.	I have worked with the NIHR RSS in conjunction with my academic supervisors to ensure that my methodologies are detailed and the analysis plans are appropriate to the research question
PPIE lacked detail	I have addressed previous issues with my use of PPIE. My work streams have changed to reflect parental and child priorities about information regarding injuries and I have detailed my PPIE methodology to date in more detail. I have also as suggested adopted a co-design PPIE methodology throughout my project proposal in coordination with CA:RING.
Missed opportunities for dissemination of research	I have changed my strategy for dissemination accordingly to include parenting groups as suggested.

Site & Support

Comments	Improvements
Inexperienced supervision from different universities	I now have two expert supervisors from the University of Cambridge with whom I have an existing relationship. Their respective backgrounds in play, injury, childhood mental health and childhood physical activity complement each other to support my PhD. I have one further supervisor who has recently moved to UCL from the University of Cambridge and has previously co-supervised PhD students with my other supervisors. She has a specific interest in adventurous play and has recently been granted a UKRI Future Leaders Fellowship for her work in active play.
Short host statement with some omissions	The host statement is revised to address specific points about research culture and the track record of supervising doctoral fellows

Training & Development

Comments	Improvements
More training required	My training plan is much more detailed with a clear vision of developing my research skills to both achieve my project proposal and to develop as a leader in epidemiology and public health.

Insufficient detail about the purpose of the overseas visit	The purpose of the overseas visit is clarified including provisional plans for a small project.

Contextual Factors
Please use this section to detail any factors that may have impacted the development of your research career or outputs from your research career to date. NIHR wants to know about any circumstances so that they may take them into consideration during the assessment of your application. Contextual factors may include:

- Career breaks due to parental leave, or periods of illness.
- Reduced time spent undertaking research activities (including research outputs), due to disability or to caring responsibilities. This may include physical or mental health related impairments or conditions, which have impacted your opportunities to undertake research activities
- Reduced opportunities for career support e.g., mentorship, and limited opportunities to undertake prior research and training.
- Impact of the COVID-19 pandemic on your research career

Please also use this section to detail any other factors that may have impacted your research career not listed in the examples provided. The impact on your career to date will be specific to your particular circumstances but could include such impacts as limited opportunities to obtain grant funding or fewer publications. In general terms, contextual factors should be significant, and relevant.

NIHR acknowledges that you may be reluctant, or uncomfortable disclosing relevant information that is sensitive. However, you should bear in mind that we are unable to take into account factors that you do not disclose. Please be assured that the information provided by you is sensitive and will be treated confidentially and in line with the General Data and Protection Regulations (GDPR).

My interest in research did not develop until late in my clinical career. I have therefore not previously applied for any of the formal academic pathways available to clinicians for which unfortunately, I am no longer eligible (AFP/ACF). This PhD was developed from my own clinical curiosity whilst working in A&E during the COVID-19 pandemic, I identified patterns in the attitudes towards physical activity from parents and clinicians following injury. This curiosity led me to independently research this area, challenging the a priori assumption that injuries are associated with adventurous play and subsequently designing this research. Without a formal research background I reached out to University departments that were conducting similar research and was adopted into the 'Behavioural Epidemiology and Interventions in Young People' team at the University of Cambridge. The team have mentored and supported me over the past three years, developing this research whilst I worked full time in paediatrics to develop an expertise in childhood injuries.

Whilst I acknowledge this lack of formal training I am proud of my commitment to learning research methodologies, my drive to embed myself into an academic environment and the outputs of this work subsequently. I have taken time out of my clinical training to pursue academic fellowships and whilst in full time clinical roles have dedicated my free time to pursuing research and developing new skills. These efforts have been recognised both by the University of Cambridge and by the RCPCH and BSHPCH with research prizes. This has had a significant affect on my career trajectory, giving purpose to my clinical work whilst developing a longer term strategy for an academic career. My next clinical rotation is a full time public health role demonstrating my commitment to this research, developing my skills and future career priorities.

I have responded to the NIHR feedback for my previous submission and I am pleased with the changes that I have made to the project. I have also worked with the NIHR RSS alongside my supervisors to ensure that the methodologies are robust and appropriate to answer the research question. I am looking forward to embarking on a clinical academic career and I believe that my strong clinical foundation give me a unique perspective to the research environment I am entering.

4. Plain English Summary of Research

Background

Playing adventurously is the most common way that children exercise. Adventurous play (AP) includes activities like climbing up trees or cycling really fast. Not only do children enjoy playing like this but it is good for their health. Children who play adventurously are more active which is really important for a healthy childhood. AP can also teach children how to manage future risks and help them develop skills which are important for when they grow up. It also improves children's mental health and can help anxiety.

Children now spend less time playing adventurously; this is bad for their health. I have shown in my previous work that this is because there are worries about them being hurt. However, there is no research that shows that children who play adventurously are more likely to be injured. In fact no recent research has been done to look at why children get hurt in the UK or how to stop it. This means we are stopping children from having a fun and healthy childhood for no good reason.

Aims

This research aims to find out whether children are more likely to hurt themselves during AP in comparison to other activities. It will also find out the reasons for children hurting themselves and what makes this more likely. It will also look at the numbers of children who have very bad injuries and work out what make this more likely.

Methods

I will do three different projects to answer the research question.

1. I will ask parents about the way their children play, the injuries their children had, how they happened, and whether children who play adventurously are more likely to be hurt.
2. I will gather information from children in Accident & Emergency departments about how they were hurt, what activities they were doing and what may have caused it.
3. I will use existing information that has been collected about serious and life changing injuries to see if there are common reasons and to see if AP is a major cause of these bad injuries.

Patient and Public Involvement and Engagement

I have talked to children, parents, doctors, nurses and charities to understand what they think about AP. Children don't seem to worry about hurting themselves, but would like information about how safe certain activities are. Parents are much more worried, about their children being hurt and about getting in trouble and blamed for their children's injuries. They have asked for me to also look at who is more likely to get hurt and specifically the effect of supervision, safety equipment like helmets and medical problems like autism and ADHD. Parents are also worried about really bad injuries which is the reason for the third project. The final details of the research will be written with parents and children to make sure the information is clear and relevant to their lives. I will work with a specific group at the University that specialise in this type of work called CA:RING.

Results

The project will bring together this knowledge to help children, parents, childcare workers, teachers and healthcare professionals have a healthy, fun and safe childhood. The results will be fed back to the people who have answered questions as part of the research, parenting groups, schools, nurseries, healthcare departments as well as other researchers at scientific conferences. It is my hope that we can learn how to safely encourage children to play as much as possible, improving children's health. Following this, I would like to use this work to look at ways to prioritise play and learn more about children's injuries.

5. Scientific Abstract

Background

Adventurous play (AP) is the predominant means by which children undertake physical activity and exercise. AP is typically defined as play, which is often outdoors, where children experience subjective feelings of thrill and excitement. It is vital for children's physical and mental health and a significant facilitator of childhood development . In my preliminary research I identified the largest barrier to engaging in AP is the perceived risk of physical injury however there is currently a significant data gap to investigate the association of injury with AP. This project will investigate the link between AP and injuries thereby addressing the largest barrier to children's physical activity.

Aims and Objectives

The aim of this research is to understand the epidemiology of AP-related injuries in children aged 0-11. The objectives are:

1. To determine the population prevalence of AP-related injuries, the circumstances in which these occur and investigate the relative risk of children suffering an injury in those who engage in high levels of AP.
2. To investigate the aetiology of childhood injury presenting to hospital, the risk factors for injuries and the role of AP in these injuries.
3. To evaluate the aetiology of major trauma in children aged 0-11 in England and Wales and the role of AP in significant traumatic events.

Methods

Objective 1: Population-Level Survey - A cross-sectional national survey of parents with children aged 0-11. The survey will collect comprehensive data on the prevalence and nature of AP and related injuries and associated factors. It will determine the relative risk of injury in children who engage in high levels of AP.

Objective 2: Detailed Hospital-Level Data Collection - A prospective cohort study will be conducted at Addenbrooke's Hospital collecting detailed data on children aged 0-11 who have sustained injuries to compare the aetiology of injuries requiring medical attention and the healthcare burden of AP.

Objective 3: National Major Trauma Data Collection - A retrospective cohort study will analyse data from the National Major Trauma Registry (NMTR), focusing on major trauma incidents among children aged 0-11. Data will include incident details, demographics, injury specifics and healthcare details. This work stream leverages existing comprehensive data to investigate the incidence and impact of AP-related major trauma.

PPIE

Patient and public involvement and experience (PPIE) has shaped this proposal, identifying specific areas of concern for families and shaping the data analysis priorities. It also led to the introduction of objective 3 which specifically focuses on life changing injuries and the contributing factors. I plan to use the CA:RING PPIE group to co-design the details of these specific work streams throughout my PhD.

Anticipated Impact and Dissemination

The results will be disseminated through existing links between the MRC epidemiology unit and PEDAL departments with nationwide parenting, education and healthcare groups. It will also be presented at scientific conferences and through peer reviewed publications. The findings will empower children, parents, teachers and childcare workers to make informed decisions about allowing AP. It will also inform national policy and public health measures to prioritise children's wellbeing.

6. Detailed Research Plan

1. What is the problem being addressed?

Adventurous play (AP) is the most common form of physical activity for children outside of school and children get more exercise from play than from time spent at clubs and organised activities.[1] AP is defined as play involving thrill, fear and risk.[2] There are 6 types of AP.[3]

1. Play at heights
2. Play with speed or situations involving uncontrolled speed and pace
3. Play with dangerous tools
4. Play near dangerous elements e.g. water
5. Rough-and-tumble play e.g. play-fighting
6. Play where children can disappear/'get lost' or be out of the sight of adults.

There is an increasing body of evidence supporting the fact that AP has significant health benefits. These benefits include:

- Increased physical activity and better physical health[4]
- Better mental health and more resilience[5]
- Improved motor skills and child development benefits[6]
- Risk assessment and injury prevention[6]
- Childhood independence[6]

Children in 2025 are engaging less in AP than children of previous generations.[7,8] Concurrently, there have been nationwide increases in the prevalence of childhood obesity and mental health disorders.[9,10] There is a direct link between AP and these health outcomes.

My preliminary research conducted with my proposed supervisors has found that the biggest barrier to children engaging in AP are concerns regarding safety and risk of injury[11]. These are similar to the reported barriers for primary school aged children by parents and by teachers.[12,13]

Currently there is no data or previously conducted research to suggest that AP is significantly associated with injuries. The unsubstantiated concerns regarding AP and injury have led to a restrictive health and safety focus and has directly contributed to a societal intolerance of AP.[14]

Concerns about injuries more generally however, are relevant. Accidental deaths caused by injuries remain the second leading cause of death in children aged 1-9 in the UK and injuries represent a considerable healthcare burden to the NHS.[15].There has never been any research investigating or demonstrating an association between AP and increased risk of injury. However there also isn't any detailed injury data collection in the UK to allow this to be researched.

This PhD proposal seeks to address this evidence gap, particularly in the UK context, by rigorously investigating the link between AP and traumatic injury in children aged 0-11 years. Addressing this barrier requires high quality injury data that is granular enough to address these concerns regarding children's safety.

What is the impact on patients, the public and health services?

First and foremost, children have a fundamental right to play.[16] This should be protected by society, not restricted. When children play, they have a natural propensity to play adventurously and therefore it is urgent and necessary that we aim to facilitate this, whilst protecting them from significant harm. This begins by studying and

understanding this relationship.

Secondly, there is a clear link between AP and children's wellbeing. Informing public health interventions to safely promote AP will address the deterioration in children's mental and physical health. AP is linked to playing outside, addressing barriers to engaging in AP will facilitate children to play outside independently which is a current NIHR research priority.[17] AP has also been identified as an important factor by the NIHR in promoting child physical activity, mental health and family wellbeing.[18]

Finally, this research will inform, assist and empower professionals involved in childcare, education and healthcare in the following ways.

1. Paediatric Emergency Medicine - a better understanding of AP and the relationship with injury will facilitate clinician's interactions with families in paediatric emergency departments. Often in paediatrics we are asked, how do we stop this from happening again and should the child be prevented from engaging in certain activities? There is no current consensus as there is no evidence informing these decisions leaving the advice at the discretion of the individual.

2. Safeguarding – Injuries are a huge concern to social workers, health visitors, and community paediatricians. Current health visiting resources do not mention the benefit of positive risk taking and only emphasise the importance of strict supervision. Current policies demand that a safeguarding referral is made for children who suffer significant injuries whilst playing, unless they were sustained during sporting activity. This work could help counter societal intolerance toward play related injuries and in turn reduce unnecessary referrals to an already overstretched social care service.

3. Teachers and childcare workers - current evidence suggests that teachers' fears of litigation relating to children's injuries stifle their ability to encourage free outdoor play.[19] This research could empower them to encourage more active play

2. Why is this research important in terms of improving the health and wellbeing of the public?

AP and physical health

AP is not only a fun pastime for children, but activities that involve risk and adventure also have positive health outcomes. Children have a natural propensity to engage in AP, regardless of perceptions of the associated dangers.[20] In a large study of Norwegian children, one of the largest proportions of observed free play was categorised as AP indicating that children prefer to play in this manner. AP is also positively correlated with increased levels of physical activity, increased enjoyment, and increased involvement in their play.[4] The associated adventure of AP is often associated with higher intensity physical activity outdoors. Therefore, limiting children's play can be linked to the rise in childhood obesity and poor health. 25% of children aged 4-5 in the United Kingdom were classified as overweight or obese in 2018/19 and 38% of 10-11 year olds in 2020/21.[21]

AP and development/mental health

AP has developmental benefits, testing physical and psychological limits and developing gross motor skills. Risk taking behaviour also encourages children to develop individual tolerance thresholds, creating boundaries to prevent injury throughout childhood.[20]

Furthermore, nationally representative data from parents of 5-11 year-olds suggests an association between higher levels of AP and fewer symptoms of anxiety and a higher affect. This association appears to be particularly strong in children from socioeconomically disadvantaged backgrounds, suggesting that these children in particular may benefit both mentally and physically by engaging in greater AP.[5]

In summary, AP is linked to better physical health, better mental health outcomes, better risk analysis and therefore theoretically fewer injuries and improved developmental outcomes. These benefits are particularly seen in children from socioeconomically disadvantaged backgrounds.

Childhood Injuries

Childhood injuries are a significant public health concern. Annually, tens of thousands of hospital admissions for non-intentional injuries are recorded in the UK, with 45,661 admissions for children under five in England and Wales in 2022/23 alone.[22] These injuries not only cause immediate harm but also lead to long-term health consequences and substantial costs for health services, local authorities, and families. Moreover, injury rates are

higher in socioeconomically deprived areas indicating a clear link between health inequalities and injury prevalence. Families in deprived areas often face greater risks due to poorer housing conditions and limited access to safety education and equipment. Addressing these disparities is crucial for reducing injury rates and promoting child safety across all communities.

Very little research is dedicated to investigating injuries in childhood. Current National Institute for Health and Care Excellence (NICE) guidance for 'Unintentional injuries and prevention strategies for under-15s', was published in 2010 and reviewed in 2019. The 2019 review found that there was no new evidence to impact guideline recommendations. It also contains unactioned recommendations to improve data quality relating to injuries in children, including the establishment of a national injury database. This NICE guidance specifically mentions the notion of countering excessive risk aversion and promoting children developing skills to assess and manage risks. My primary supervisor is a key contributor to this NICE guidance and is specifically interested in the role of injury and risk analysis.[23] My proposed research would contribute directly to this call for evidence.

3. Review of existing evidence - How does the existing literature support this proposal?

Despite the association between AP and positive health outcomes in children, there is a paucity of existing data examining the association between AP and the most commonly cited barrier; injury. Injury statistics in paediatric studies are often categorised by place of accident rather than activity[25]. To date no studies have collected data examining AP related injuries.

Studies investigating the barriers to AP are dominated by concerns about physical injury and safety. Four out of the top five barriers identified in the British children's play survey (children aged 5-11) were related to safety.[12] This was corroborated in the British pre-school play survey (children aged 0-4) again with the top two barriers cited as safety and risk of injury.[11] These concerns are replicated in international data from Canada, New Zealand, Australia and the USA as the most significant influence on children's access to independent play for parents, teachers and childcare workers.[13,14,19,20]

A key recommendation emerging from the literature is the urgent need for epidemiological studies focused specifically on play-related injuries to be able to effectively mitigate parental concerns regarding safety and the risk of injury.[11,26] Such studies would provide a robust foundation for identifying patterns, trends, and risk factors associated with injuries during AP. Parents who view risky play as beneficial are more likely to encourage their children to engage in it.[20] Furthermore, epidemiological studies would enable the development of targeted interventions aimed at mitigating injury risks whilst promoting AP. The appetite for this is reflected by data from research that found that parents largely supported a reduction in health and safety policies and would be supportive of encouraging children to engage in more AP, if they felt it was safe.[27,28]

4. What is the research question / aims and objectives?

Research Question: What is the association between AP & injuries in children aged 0-11?

The overarching aim of this proposal is to comprehensively understand the epidemiology and societal impact of AP-related injuries in children aged 0-11. This age range has been chosen due to the importance of AP in this specific age group as a source of physical activity, in comparison to older age groups who participate predominantly in organised sport.

By exploring the prevalence, contributing factors, and relative risk associated with these injuries the study seeks to inform evidence-based interventions and policies that promote AP whilst preventing injuries; ultimately enhancing child health and development.

Objectives:

1. To determine the population prevalence of AP-related injuries, the circumstances in which these occur and investigate the relative risk of children suffering an injury in those who engage in high levels of AP.
2. To investigate the aetiology of childhood injury presenting to hospital, the risk factors for injuries and the role of AP in these injuries.
3. To evaluate the aetiology of major trauma in children aged 0-11 in England and Wales and the role of AP in significant traumatic events.

5. Project Plan

The above objectives will form the following three workstreams (WS) to comprehensively address the research question. They have been formulated alongside preliminary PPIE as detailed in the relevant section. The use of PPIE throughout the project plan refers to the use of the Children and Adolescents: Research Involvement for the Next Generation (CA:RING) service.[29] This PPIE resource is facilitated by the University of Cambridge (UoC) and supported by the NIHR. Further details can be found in the PPIE section of this proposal.

To understand the epidemiology of AP and injury, and therefore address this barrier to physical activity, the project proposes to investigate this relationship at the population level, in detailed healthcare data collection and nationwide trauma statistics.

WS1

Objective: To determine the population prevalence of AP-related injuries, the circumstances in which these occur and investigate the relative risk of children suffering an injury in those who engage in high levels of AP.

Introduction

It is not possible to ascertain the risk of adventurous play by only collecting data on children who are already injured. Episodes of AP are numerous in the community and the majority of these do not end in visits to A&E. It is therefore imperative that a population level study be conducted to analyse both the prevalence of AP related injuries but also to ascertain whether there is a correlation between increased risk of injury with increasing AP exposure. Analysis at the population level will also be able to identify associations between demographics, different subcategories of AP and PPIE priorities such as neurodiversity, supervision and safety equipment.

Study design

WS1 employs a cross-sectional, national survey collecting data about exposure to AP, injuries and demographics. This design and methodology is modelled on the British Children's Play survey (BCPS) and the British pre-school children's (BPCPS) survey which Dr Hesketh (named supervisor) developed.[7,8]

These previous studies focussed primarily on access and correlations between AP frequency and demographics but not on injuries and the circumstances in which they occur. The survey will be redesigned with expert guidance and PPIE input.

Recruitment

Participants will be recruited through YouGov, a large online platform which collects data for large national surveys, polls and market research. YouGov uses active sampling to ensure a nationally representative sample and it's established platform reaches a broad range of socio-economic, ethnic, and educational backgrounds, ensuring equity, diversity and inclusion (EDI) of participation. Participants will not receive direct payment however will accrue YouGov points, a system which can be redeemed for rewards and in-turn cash payment. This recruitment method has previously been used successfully for the BCPS and BPCPS and efficiently gathers national and representative data inclusive of key EDI demographics.

Data collection

Sample size

Using an AP injury prevalence of 6% as determined by pilot data collected in my preliminary research and a 95% confidence level, the required sample size is approximately 1,100 participants after adjusting for a 10% non-response rate. To allow subgroup analyses, and address potential non-response, we propose a total sample size of 1,500 participants**.** This approach will provide reliable estimates of prevalence and risk ratios, supporting the study's objectives.

Data to be collected

Participants will complete the survey online via the YouGov platform. The survey will be co-designed through PPIE to ensure clarity, accessibility, and relevance. We will also aim for the questions to be non-judgemental to avoid responder bias. All responses will be anonymous.

Survey Content:

- Demographics
    - Age, gender, socio-economic status, rural/urban living

    - PPIE priorities such as neurodiversity/co-morbidities
- AP Frequency and Exposure
    - Frequency and duration of AP.
    - Types of AP (e.g., play at heights, rough-and-tumble play).
- Injuries:
    - Type, severity, and frequency of injuries.
    - Context of injury (e.g., supervision levels, safety equipment use).

Data analysis

The focus on relative risk is an important theme that runs through the project and was guided by PPIE as parents found it more useful to have an idea of the risk in comparison to other activities/other groups rather than the absolute risk.

- Descriptive Analysis:
    - Exposure to AP (frequency, intensity) – methodology similar to BCPS/BPCPS[7,8]
    - Prevalence and characteristics of AP-related injuries.
- Risk Analysis:
    - Calculate relative risk and odds ratio of injuries in children who have a significant exposure to AP versus children with low levels. Analysis will be dependent on reported AP exposure but will aim to identify correlations between AP and injury.
    - Multi-level regression modelling to adjust for potential confounders (e.g., socio-economic status, neurodiversity).
- Subgroup Analysis:
    - Analysis of factors such as supervision, safety equipment, and specific AP types. (*PPIE priority)* Comparison between demographic groups i.e gender and injury rates as secondary analyses.

Conclusion

WS1 will investigate population level correlates between AP exposure and injury, identifying those children who engage in AP without sustaining injury and those injuries that don't present to healthcare settings but may influence behaviours. By investigating the exposure of children to AP, associated injury risks, and contributing factors, this research will address key evidence gaps and inform public health strategies to promote safe, adventurous play for children.

WS2

Objective: To investigate the aetiology of childhood injury presenting to hospital, the risk factors for injuries and the role of AP in these injuries.

Introduction

Whilst injury remains the largest barrier to AP, preliminary data indicates that AP related injuries may be relatively uncommon. Therefore, it is important to look at A&E presentations to understand the relative burden of AP related injuries in comparison to other aetiologies. During my predoctoral fellowship I found that existing emergency and hospital databases lack granular detail regarding aetiologies of injuries and are inadequate to address concerns regarding injury circumstances. Detailed hospital level data is essential at comparing the types of injuries sustained during AP in comparison to other aetiologies, the associated burden and the circumstances in which these more serious injuries occur.

Study Design

This project employs a prospective cohort study design to collect detailed hospital-level data from A&E in children who have sustained injuries. A pilot study conducted in July 2024 revealed significant gaps in the level of detail typically found in A&E clerking to answer the research question, necessitating a shift from a retrospective to a prospective data collection method. The proposed centre benefits from having a minor injuries unit and an A&E as well as being a major trauma centre for the East of England with a high volume of paediatric emergency cases, ensuring a diverse and representative sample of patients, injury types and outcomes. Children admitted or requiring outpatient follow-up i.e fracture clinic will be followed-up via the electronic health records system at Addenbrooke's hospital (EPIC).

Recruitment

All children attending A&E with an injury for a one year period – estimated to be between October 2026 and October 2027, will be approached for recruitment. Given the prospective nature of the study, the sample size will

be determined by the number of eligible cases that present to A&E. Based on historical admissions data, it is estimated that approximately 500 cases will be reviewed.

Data Collection

Data will be collected from patient records using the EPIC system. To minimise the discrepancy in quality between data collection, a pro forma will be developed. The exact detail of these pro formas and data collection will draw on previous validated injury surveillance methodology and be tailored through an iterative approach using additional PPIE focus groups. Additional training will be delivered to staff in A&E about this pro forma to ensure detailed data continues to be collected and the PI will additionally participate in data collection where possible. Note that my training plan includes clinical time spent in the A&E department which will allow me to troubleshoot frequently. Addenbrooke's hospital uses the EPIC system which allows 'smartphrases' to be built and shared amongst all clinical staff. A smartphrase will be developed embedding the pro forma so that clinical staff can easily input data when the principal investigator is not collecting the data. Hospital records will be later accessed for those children suffering substantial injuries that require follow-up/admission to hospital.

Data to be collected includes:

- Demographics: Age, gender, socio-economic status, ethnicity.
- Injury Details: Type, severity, context, supervision (direct/indirect), and use of safety equipment. If injuries occurred during AP, subcategorisation into the specific type of AP will be collected.
- Diagnoses and Treatment Plans: Detailed notes on diagnosis, treatment, and follow-up
- Co-Morbidities: Presence of any co-existing medical conditions. PPIE highlighted the interest in ADHD and autism.

Data Analysis Plan

Descriptive statistics will be used to summarise patient characteristics, types of AP, types of injury, and circumstances of injury. The sample size will be assessed to check for appropriate powering of comparative analysis which could include comparing the severity/frequency of injury with:

- Different activities/categories of AP
- Demographics and co-morbidities (PPIE priority: neurodiversity)
- Supervision/safety equipment (PPIE priority)

Logistic regression models will identify any significant correlation between variables and injuries. Independent variables will include AP subcategories, age, gender, SES, supervision levels, and safety equipment usage. This will allow for the identification of risk factors that increase the likelihood of injuries occurring during AP.

Summary

By collecting and analysing data about children who have sustained injuries, the relative burden of AP-related injuries in comparison to other aetiologies, such as sports and non-accidental injury, can be ascertained. This is essential in addressing parental concerns about the relative safety of AP and puts the cause of these injuries into context.

WS3

Objective: To evaluate the aetiology of major trauma in children aged 0-11 in England and Wales and the role of AP in significant traumatic events.

Introduction

WS3 is a direct response to PPIE, stemming from parental concerns about serious injuries that arise from play and to understand the specific circumstances in which they happen. Major trauma in the UK is defined as isolated or multiple serious injuries that may result in death or disability and is the focus of this work stream.

Study Design

This study is a retrospective cohort study design analysing data from the National Major Trauma Registry (NMTR), a national database maintained by NHS England.[30] The registry includes comprehensive trauma data from hospitals across England, providing a rich source of information on major trauma incidents. The focus is on assessing the incidence of AP-related major trauma among children aged 0-11. This project will additionally attempt to identify specific areas of injury prevention if there are common themes identified in AP related major trauma. This database has been accessed prior to this proposal submission and has comprehensive data for injuries that meet the threshold of 'major trauma'.

Sample Size

The sample size is determined by the availability of NMTR data for children aged 0-11. Data from 2022-2025 will be used as this data is less confounded by a change in behaviour necessitated by the COVID-19 pandemic. The registry's comprehensive nature ensures a large sample, which is expected to be statistically robust. The most recent NMTR data report (Jan 2019-Dec 2020), there were approximately 1100 children 0-11yrs with major trauma in England and Wales.

Data Analysis Plan

All cases of childhood major trauma will be analysed. A coding mechanism for all aetiologies of major trauma will be developed and a 'code' will be assigned to each case to allow aggregation of common causes for comparative analysis. Data will be extracted from NMTR records and analysed using descriptive methods to understand the prevalence, severity, and outcomes of national AP-related major trauma. It is anticipated that AP will not contribute highly to major trauma and therefore the current plan is for all of these cases to be identified and analysed in granular detail to identify common themes that can be targeted in measures for prevention. If this is not feasible a broader approach will be taken. The analysis will then put AP-related trauma into the context of other causes using comparative and regression statistics to compare between the subcategories of AP, demographics and non-AP related aetiologies of trauma where sample size allows.

Summary

WS3 will address priorities informed by PPIE regarding significant and life changing injuries. WS3 will provide detailed analysis of AP-related major trauma cases aiming to identify possibilities for prevention. It will further put AP-related injuries in the context of other aetiologies of major trauma further addressing the research question from a national database.

Overall Project Conclusion

Together, these three work streams create a comprehensive overview of the relationship between injury and AP. The population-level survey investigates the relative risk of AP on injury rates and encompasses children who have not sought healthcare advice for their injuries and importantly surveys children engaging in AP who have not sustained injuries. The hospital-level data provides detailed insights into the healthcare burden of injuries requiring medical attention and the factors that contribute to this. Finally, the national trauma data contextualises the most severe outcomes within the broader landscape of childhood injuries. This multi-tiered method investigated parental concerns into the largest barrier to AP.

6. Dissemination, Outputs and anticipated Impact

What do you think the impact of your research will be and for whom?

This research will directly address the biggest barrier to physical activity in children. It is therefore imperative that is disseminated to children, parents and the relevant professionals that interact with children including education, childcare and healthcare. The MRC epidemiology unit and PEDAL unit both have existing infrastructures, through both local and national parenting groups, public health bodies and national policy makers whereby this information can be directly transmitted in order to shape public health strategies.

I wish for this research to:

- Improve children's access to AP, outdoor play, and physical activity
- Reduce the number of significant injuries sustained by children
- Contribute to further research and broader public health policy

What do you intend to produce from your research and how will your outputs enter society as a whole?

- Research outputs - Academic publications in peer-reviewed journals and presentations at national and international conferences. This will provide a peer-reviewed evidence base which may inform injury prevention as well as public health strategy
- Generate a new comprehensive dataset that captures the prevalence, causes, and outcomes of AP-related injuries in children aged 0-11 in the UK. This dataset will address the current gap in detailed injury data and will be made available for future researchers through the UK Data Service, following relevant embargoes.

- Educational materials: Development of resources for use by parents or workers in schools and nurseries to promote the benefits of AP whilst outlining the evidence-based findings of the associated risks.
    - Parents and educators will be targeted with additional resources through parenting groups and educational networks
    - For academics and healthcare workers these will be disseminated through existing and new relationships within healthcare, publications and presentations.
- Policy recommendations aimed at influencing national public health strategies and safety regulations concerning children's play environments.
    - The PEDAL unit at the UoC (led by supervisor Professor Ramchandani) have experience with policy formulation and implementation which I aim to learn from dependent on the outcomes of the research. Through this I will have access to policy makers.

Future Research

Future work for this research will firstly be to directly inform measures to prevent injuries by detailing the aetiologies and exact mechanisms of AP related injuries. It will also help us contextualise AP as a risk factor for injury in comparison to other common childhood activities, to help parents, childcare professionals and policy makers to take informed decisions about the activities they promote.

Ultimately, this research is positioned to have a lasting impact on child health and development, public health policy, and the overall wellbeing of families and communities across the UK.

7. Project Management

The project will be managed in close collaboration with my supervisors through regular scheduled meetings to ensure progress aligns with the research timetable. Key milestones have been established to track progress (please see research timetable) and allow for timely completion of each workstream. Data access and ethical approvals will be prioritised early, with support from the MRC Epidemiology Unit to streamline the process.

I will ensure all data collection follows validated methodologies, with adjustments informed by PPIE workshops and expert guidance to maintain robustness. Data will be securely stored in compliance with NHS information governance standards and UoC protocols. Pilot studies will be undertaken where necessary to refine data collection tools.

Support for insurance, sponsorship, and CRN approvals will be secured through the UoC to ensure operational readiness. This structured management approach will ensure the project remains on schedule and delivers high-quality outputs.

8. Ethics

The following ethical considerations have been made in conjunction with MRC and HRA guidelines following discussions with Caldicott guardians. All researchers will be compliant with NHS GCP standards as per HRA guidance. Ethical approval will be sought from the appropriate research ethics committee at all stages. All data will be securely stored and handled in accordance with the General Data Protection Regulation (GDPR), with the MRC Epidemiology Unit providing expert oversight on data governance. UoC ethical approval will be sought for all workstreams.

WS1

- All survey responses will be anonymised, and any personally identifiable information will be strictly avoided. YouGov follows robust ethical practices and adheres to strict data protection guidelines.

WS2

- Approval will be sought for the prospective collection of data from Addenbrooke's Hospital. This will be subject to both University and Hospital approval and work to understand these processes locally has been undertaken.

WS3

- The use of retrospective data from the National Major Trauma Registry (NMTR) requires section 251 approval under the Health and Social Care Act for the analysis of pseudonymised data. Approval for secondary use of the data will be obtained.

9. Success Criteria

Measurements of success

- Presentations at several stages of this research at key conferences sharing knowledge and presenting data.
- Development of high quality methodologies using PPIE - particularly in the co-design of the population-level survey and dissemination of results to parents and caregivers.
- Successful recruitment of study participants within the appropriate timeframe
- At least three publications in a peer reviewed journal.

If AP is not seen to be significantly associated with increased risk of injury an ambitious goal of this research is to develop a statement alongside key stakeholders such as the RCPCH, the Royal Society for the Prevention of accidents with a balanced message supporting AP. This has been done by the Canadian Paediatric Society unfortunately without the requisite supportive data about injury risk.

Key Risks and Contingencies

WS1 Risk: Low Survey/Recruitment Rate
Contingency: Using YouGov's platform, which is known for its wide reach, and collaborating with hospital clinicians will ensure strong recruitment. A flexible timeline allows for extension if needed. The following workstreams are not dependent on these results.

WS2 Risk: Poor quality data from A&E

Contingency: I have built time into my plan to allow adequate education and the development of a pro forma to ensure that key data will be obtained. If this remains inadequate adjustments to the analysis will have to be made and alternative routes for data collection i.e the PI will collect data for one week per month to allow for seasonal variation and work with a lower sample size.

WS3 Risk: Delays in Data Collection Access
Contingency*:* I have pre-secured access to key datasets and established strong institutional connections. Regular communication with data providers will ensure clear timelines. In case of delays, I will explore alternative databases such as PICAnet which has data pertaining to paediatric intensive care unit admissions, some of which are trauma related.

7. Patient & Public Involvement

Please describe how patients/service users, carers and the public have been involved in developing this proposal

During the development of this proposal, I have actively involved children and their caregivers through various stages of engagement to ensure the research is directly aligned with their concerns and priorities.

During my predoctoral fellowship, I engaged in opportunistic conversations with 19 willing parents of children who had sustained injuries. These informal discussions provided early insight into parental concerns around AP. Caregivers expressed a desire to understand the balance between encouraging AP and preventing injuries, especially in regard to whether certain activities were too dangerous. These discussions shaped the initial research focus, emphasising the need to understand the relationship between AP and injury prevention, while considering the impact on children's mental and physical development. Parents and children wanted to know how common injuries are in AP and whether particular activities, like climbing or high-speed play, pose higher risks. Children understood the need to be safe but wished to have more freedom to make their own mistakes even if that meant being injured. This informed Workstream 1.

Building on these insights, I worked with the PPIE team at Guy's and St.Thomas' Hospitals Trust to engage in further discussions. A group discussion with a diverse group of 10 parents, recruited at the Mary Sheridan children's community centre in Lambeth helped shape further details of this work. This is summarised here:

- Factors contributing to injury: Parents were particularly interested in understanding what factors beyond the activity itself—such as supervision levels, use of safety equipment, and neurodiversity (e.g. attention deficit hyperactivity disorder or autistic spectrum disorders)—contribute to injury risk. This shaped the data collection strategies for WS2&3.
- Significant or life-changing injuries: There was a strong desire to investigate major, life-altering injuries. My previous proposal did not include this consideration and therefore Workstream 3 was included.

This feedback has been instrumental in shaping the focus of the research proposal, particularly in addressing parental concerns around injury risks and factors such as neurodiversity, supervision and safety equipment. Through this engagement, I have ensured that the research addresses the real-world questions and priorities of parents, enhancing its relevance and impact on public health and safety.

Please describe the ways in which patients/service users, carers and the public will be actively involved in the proposed research, including any training and support provided

PPIE approach, management, and support

Approach

Strategy for this project prioritises the integration of insights from families and young people. The University of Cambridge has access to a bespoke PPIE service for children and adolescents (CA:RING) which provides tailored support to incorporate PPIE throughout research projects and dissemination at the University.

Management

CA:RING has a dedicated co-ordinator with access to a diverse group of over 200 PPIE participants. Initial orientation meetings have taken place with a broad PPIE strategy agreed upon. CA:RING uses two national groups of participants a parent group and a young people group. National representation ensures a more diverse subset of participants and CA:RING actively targets underrepresented groups to ensure EDI. Throughout the project, I will work with facilitators to support the coordination of activities, such as co-designing survey questions and providing feedback on study materials. These interactions will be documented and incorporated into research adjustments to ensure ongoing alignment with the priorities of families and young people.

Summary of PPIE Activities

WS1: Population-Level Survey: PPIE contributors will co-design survey questions in a focus group session, ensuring these reflect parental concerns about injury risks. Feedback from parents during survey testing will refine question phrasing to enhance clarity and relevance.

WS2: Hospital Data Collection: Co-development of a standardised pro forma for data collection through a focus group session, ensuring this captures key variables. Information sheets and consent processes will be reviewed separately by participants.

Workstream 3: NMTR Data Analysis: Parents and young people will guide data analysis and priorities within the data set from the trauma data, contextualising the types of injuries and adventurous play scenarios that are most concerning to families and what is the most valuable information for them to learn.

Dissemination: PPIE members will be engaged in developing and reviewing draft guidelines and educational materials developed from the study's findings, ensuring they are user-friendly and address the practical needs of families. Their input will shape how results are communicated in public forums and local community events.

If it is considered not appropriate and meaningful to actively involve patients/service users, carers and the public in your proposed research, please justify why
Patient and public involvement is key to this proposal and I have endeavoured to involve children and their parents at all stages of this proposal. I will continue to do this throughout my fellowship and will undertake specific training to incorporate this in my future academic career ensuring EDI representation throughout.

8. Training & Development and Research Support

Proposed training and development programme

The following training plan has been developed with my supervisors and aims to develop my practical research skills and leadership capabilities whilst factoring in my career aspirations.

University of Cambridge training

The University of Cambridge offers a robust range of training opportunities for PhD students. I will have access to world-class resources and expert-led training through its various departments and research centres. I will undertake courses within the department on advanced statistical analysis, epidemiological study design, and data management tailored for large-scale population health research. The MRC epidemiology unit benefits from weekly seminars, a strong graduate community and guest lecturers from world leaders in their respective fields.

Equality Diversity and Inclusion

Additionally, the Cambridge Centre for Equality Diversity and Inclusion offer the following professional development workshops in which I will enrol:

- Equity, Diversity and Inclusion course - the main principles of equality and diversity and legislation and how equality and diversity impacts on research
- Understanding unconscious / implicit bias and the effect on research

Data Management and Research Techniques

- **Epidemiology Short courses**
-
    - London School of Hygiene and Tropical Medicine
    - 15 days total over 2 years
    - To ensure rigorous data collection and analysis, I will complete courses focusing on handling complex datasets, epidemiological analysis, and public health surveillance methods. These skills are directly applicable to the analysis of national survey data, hospital-level data, and major trauma datasets.
- **Training in PPIE and Co-Development**
-
    - **University of Cambridge**
    - **Short courses – 1-3 days**
    - Engaged Researcher training covers all areas of public engagement with research by combining expert led practical workshops and taught sessions.
    - Sessions will teach me how to develop, deliver, communicate, publish and evaluate your Public Engagement activities with a focus on EDI and representation of under reported groups.

Workstream 1: Population-Level Survey

1. **Training in Data Management and Statistical Analysis**
    - **University of Oxford**
    - **Short course 3 days**
    - Effective data management and statistical tools are essential for handling the large dataset expected from the survey. This training will cover data cleaning, coding, and structuring techniques required for accurate and reliable analysis.

Workstream 2: Detailed Hospital-Level Data Collection

1. **Course on Advanced Epidemiological Analysis**
    - **Institution:** LSHTM
    - **Relevance:** This advanced course will enhance my epidemiological skills, focusing on regression modeling and risk assessment necessary to analyse the relationship between adventurous play and injury outcomes in hospital settings.

Workstream 3: National Major Trauma Data Collection

1. **Big Data Epidemiology**
    - **Institution:** University of Oxford
    - Data linkage techniques are essential for integrating trauma data with other sources. This training will enable more comprehensive analysis of contextual factors associated with injuries, enhancing the depth and applicability of findings in WS3.

Additional Courses and Training

Postgraduate Certificate in Teaching and Learning in Higher Education

-

- **Institution:** University of Cambridge, Institute of Continuing Education
- 5 days per year + coursework
- To prepare for future academic roles, I will enrol in this programme available through the **University of Cambridge's Institute of Continuing Education**. This course will strengthen my teaching skills, allowing me to deliver engaging and impactful lectures to students and wider audiences. It will also support my goal of becoming an academic clinical lecturer.

Kings Fund - Leadership for Population Health

- Programme designed to support with working strategically with partners from across the health and care system to improve the health of your local population.
- The programme consists of four modules each lasting two days. There will be a mix of practical exercises, expert content, learning from case studies in UK and internationally, individual and group development.

Research Visit – University of British Columbia

-
  - **Location:** University of British Columbia, Canada
  - 6 weeks
  - This placement provides an opportunity to learn from a global leader in adventurous play research. Professor Brussoni's mentorship will enhance my skills in injury prevention strategies and research methodologies, aiding in the translation of research findings into actionable UK policy recommendations. Full details can be found in 'Collaborations'.

Beyond these formal courses, I will benefit from access to seminars, workshops, and events hosted by the Cambridge Public Health Interdisciplinary Research Centre, which bring together experts in public health and policy. This will allow me to stay updated on the latest research trends and engage in valuable interdisciplinary networking opportunities.

Policy and International Collaboration

The Play, Education Development and Learning unit at the University of Cambridge is actively involved in shaping policy as a leading hub for play-related research. Through its links to charities and policy makers I will gain insights into how to translate research into actionable policy, which will be critical for advocating evidence-based changes in how risky play is approached in the UK. The department has specific links to senior members of the civil service and contribute to policy and the Royal Foundation.

Clinical Training

I am committed to maintaining a clinical role alongside my research activities. Following discussions with my supervisors, I propose to dedicate one day per fortnight to clinical practice at Addenbrooke's Hospital in paediatric A&E. This arrangement allows me to stay engaged with direct patient care, ensuring that my research remains grounded in real-world clinical practice and that my skills as a practicing clinician remain up-to-date. This continued clinical exposure is crucial for refining the practical aspects of my research, particularly those involving patient interactions and outcomes.

Summary

This training programme, combining advanced research methods, PPIE skills, policy advocacy, and leadership development, is designed to equip me with the competencies necessary for impactful research. Through targeted courses, international collaboration, and expert mentorship, I will be prepared to advance the field of paediatric public health and contribute to the research of children's physical activity and injury prevention.

RESEARCH SUPPORT / SUPERVISOR 1

Name	Dr Esther van Sluijs

Institution	University of Cambridge

Position	Programme Leader and MRC Investigator Behavioural Epidemiology and Interventions in Young People

Description of support provided
Dr Esther van Sluijs is a Behavioural Epidemiologist with expertise in observational and intervention research

focussed on the understanding and changing young people's physical activity behaviour. She is an internationally recognised expert on young people's physical activity behaviour and has contributed to the development of NICE guidance on "Promoting Physical activity for Children and Young People", the IOC consensus statement on the "Health and Fitness of Young People Through Physical Activity and Sport", and the 2018 UK CMO Physical Activity Guidelines Update, and is currently a consultant on young people's physical activity for the World Health Organisation. She has extensive experience in the design and conduct of observational and intervention studies and is the Director of the UKCRC-registered Cambridge Epidemiology & Trials Unit.

Esther has successfully supervised 11 PhD students to completion (including two funded by NIHR School for Public Health Research (SPHR)) and is currently primary supervisor for two NIHR SPHR funded students. Esther has conducted 18 PhD examinations and mentored six early career researchers. In 2023, she was awarded an International Society for Behavioural Nutrition and Physical Activity mentoring award for her educational supervision.

Esther will be Tom's primary supervisor. She will meet with him regularly in between monthly meetings with the supervisory team, be a day-to-day contact and lead on providing feedback on written work. Tom will join a vibrant team of mixed-career researchers at the Behavioural Epidemiology and Interventions in Young People team, with bi-weekly team meetings facilitating peer support and joint learning.

RESEARCH SUPPORT / SUPERVISOR 2

Name	Professor Paul Ramchandani

Institution	University of Cambridge

Position	LEGO Professor of Play in Education, Development and Learning

Description of support provided
Paul is the LEGO Professor of Play in Education, Development and Learning at the University of Cambridge and Director of the PEDAL Research Centre. Prior to this he was Professor of Child and Adolescent Health at Imperial College London. He trained initially as a medical doctor and continues to work as a consultant in the NHS. Paul works with a team studying early child development and developing and testing interventions to promote children's mental health and children's wider positive development, with a significant focus on children's play and health. Paul is a trustee of Foundations, the What Works centre for children and families (previously the Early Intervention Foundation) and is a member of the Scientific Advisory Council for the UK Department for Education. Paul previously held training fellowships from the MRC and Wellcome Trust and has led and contributed to a range of NIHR-funded trials. He has successfully supervised 11 PhD students to completion and is currently primary supervisor of six PhD students. Paul will be Tom's second supervisor. He will meet monthly with Tom and the supervisory team and also as a member of the PEDAL Research Centre. Paul has worked with the MRC epidemiology team to promote children's adventurous play and currently co-supervises another PhD student combining the study of play and physical activity. He will also be instrumental in the dissemination and output of this work as a key stakeholder in national policy formation. Paul and Esther have previously co-supervised other PhD students.

RESEARCH SUPPORT / SUPERVISOR 3

Name	Dr Kathryn Hesketh

Institution	University College London

Position	Associate Professor of Behavioural Epidemiology

Description of support provided
Dr Kathryn Hesketh is a UKRI Future Leaders Fellow and Associate Professor of Behavioural Epidemiology in the department of Behavioural Health and Science (BHS) at UCL. She also holds a visiting fellow position at the MRC Epidemiology Unit, Cambridge. She is an expert in physical activity behaviours and risky play in the under 5s, and with training in health psychology, has wide-ranging experience of evaluating risk perceptions of injury and resulting behaviours in adults and children. She was a member of the Chief Medical Officer's Physical Activity Guidelines and Surveillance committees (for 2019 Guidelines) and is a Physical Activity lead advisor for the Institute of Health Visiting. Kathryn has successfully supervised three PhD students to completion and has provided a further three students with support for interdisciplinary PhDs at UK universities. She is currently first supervisor to two second-year students and topic adviser to a third, working closely with both Dr Van Sluijs and Prof Ramchandani on existing supervisory teams. Kathryn leads her own research group within the BHS, providing mentorship to several early career researchers. She will be topic adviser for Dr Rance's PhD, joining monthly supervisory meetings and supporting development of PhD outputs.

Collaborations
Play Education Development and Learning (PEDAL) unit collaboration with Starlight Foundation Over the past two years I have been increasingly involved in a collaboration and working relationship between the PEDAL unit at the University of Cambridge and Starlight Foundation, a leading children's charity which promotes play for health and in healthcare settings. I have worked with starlight CEO Cathy Gilman and Deputy CEO Dr Krutika Pau to shape the charity's future direction. This collaboration with the University of Cambridge, providing research insights into the world of play led by Professor Ramchandani and allowing the charity to develop the strategy for dissemination and advocacy is key to this projects success. Whilst the charity predominantly provides play support for unwell children, the charity's future direction is more broad, looking to promote play as a method of improving children's health more generally. The charity is well supported with annual expenditure over £5 million and are supportive of this work and have agreed to be involved in dissemination and promotion of its impact. I look forward to working with the charity throughout this project and developing stronger advocacy networks using third sector partners. **Overseas Research Visit to the University of British Columbia, Canada** **Collaboration with Professor Mariana Brussoni** Professor Mariana Brussoni is a world-renowned expert in adventurous play, injury prevention, and public health advocacy, leading a dedicated research centre at the University of British Columbia focused on outdoor play and children's health. She is the only person worldwide with expertise in both injury prevention and adventurous play. Collaborating with Professor Brussoni will provide an unparalleled opportunity to deepen my understanding of cutting-edge methodologies for researching adventurous play and injury risks. Notably, Professor Brussoni's successful work with Canadian policymakers, schools, and social care providers has led to substantial policy changes, including advocacy with the Canadian Paediatric Society to promote "risky play" as a component of child development. Observing and learning from her approach to engaging with educational and social care institutions will offer critical insights into addressing similar concerns within the UK context, empowering me to translate my findings into actionable policy changes effectively. **Training and Development Opportunities** This placement will offer hands-on experience with novel injury prevention strategies and advanced research methodologies that have proven influential in Canadian public health. Professor Brussoni's approach includes conducting randomised controlled trials focused on adventurous play interventions, as well as utilising simulated and virtual environments that allow for controlled observations of children's responses to various risk scenarios. Exposure to these innovative methodologies will not only strengthen my skills in designing rigorous, evidence-based studies but also help me gain practical expertise in testing behavioural responses, an aspect directly relevant to my project's focus on balancing injury prevention with the promotion of adventurous play. Learning how Professor Brussoni effectively advocated for policy change, particularly in contexts such as schools and social care, will provide valuable strategies for engaging UK stakeholders in a similar manner.

Project

I have been invited to contribute to a project that is currently underway with an academic social worker in Professor Brussoni's team who is exploring the role of social work and injury. The broad project aims involve trying to balance the serious concerns of non-accidental injuries with ensuring that vulnerable children have access to outdoor adventurous play without fear of being reprimanded by authorities. This has direct parallels with concerns raised by teachers nursery workers and healthcare professionals that have been identified in the UK.

Host Organisation Support Statement
The following statement of support must be completed by the head of your lead academic department. In order to contribute to this application, this individual must be invited to do so via the 'Participants and Signatories' section of the form.

Statement of support

As Co-Director of the University of Cambridge Institute of Metabolic Science (IMS) and Director of the Epidemiology Unit, I am writing to provide my strong support for Dr Tom Rance's NIHR Doctoral Fellowship.

He applied to and was interviewed by our PhD programme earlier this year and was unanimously found to meet the academic level to undertake PhD studies in our department. He submitted a strong PhD proposal on this topic of risky play, and gave an excellent interview, showing his knowledge of epidemiological research methods and willingness to enhance his skillset further during a PhD. He ranked joint first in the Epidemiology/Public Health Panel (the other being Genetics/Aetiology) and was offered a place to study for a PhD here, conditional on funding. He has since liaised frequently with investigators in our department and relevant collaborators over several months to put together this fellowship proposal on which he has shown excellent proactivity and independence of thought. During this time, he attended a National Play Conference at the University in his free time to refine his research ideas. In the time since his previous application, Tom has embedded himself within our department as a Visiting Researcher. He has conducted preliminary studies and contributed to the wider research of the 'Behavioural Epidemiology and Interventions in Young People' programme. Specifically, he has worked with his proposed supervisors, publishing peer-reviewed articles about the health benefits of adventurous play and researching the facilitators and barriers to adventurous play to support this research proposal. Additionally, he has contributed to a systematic review of quantitative tools for measuring adventurous play, collaborating with a wider network of researchers across the UK and internationally.

I confirm that this topic is of high relevance to the research goals of the Epidemiology Unit, and specifically the Behavioural Epidemiology programme. The proposed project builds on exciting interdisciplinary links between our department and the Faculty of Education with whom we have a successful record of shared PhD students (supported by an ESRC doctoral training programme). We are well placed to support his proposed mixed methods training in quantitative and qualitative skills, both through formal courses and by support of several investigators who undertake this combined approach, as well as offering a rich programme of wider skills training available at department, Clinical School, and University levels. Furthermore, our investigators include a number of clinician scientists and so we are well placed to host and support clinical academic trainees (currently 5/30+ PhD and postgraduate trainees) to balance the challenges of progressing in both strands of their training, as well as building synergistically on both experiences. Specifically for Thomas, we have close links with the University and Clinical departments of Paediatrics in which Professor Ken Ong holds a clinical position.

We are also well equipped to support his proposed research studies. Our department includes professional research support teams spanning, data management, statistics, IT, study co-ordination and field work teams, as well as the UKRCC-accredited Clinical Epidemiology Trials Unit. Alongside the many population-based cohort studies hosted in our department and available for his studies as he describes, our department's ISO27001 certified Secure Research Computing Platform allows us to link to and hold routine NHS and other potentially sensitive routine data needed for his studies.

The Epidemiology Unit is a world-leading centre of research excellence with an annual research grant income of £13.5 million per annum. It currently employs 203 staff and students. In the REF 2021, the Unit contributed 31 FTE individuals. 97% of all outputs returned by the University of Cambridge in UoA2 were rated world leading or internationally competitive. The Unit has a strong track record of producing research that informs public health policy and practice in the UK. The Unit is a member of the NIHR School for Public Health Research and since 2018 has had 9 NIHR funded students.

The Epidemiology Unit has established initiatives to support early career researchers (ECRs) with professional development, career progression, and research funding acquisition. ECRs are encouraged to share their work and

future research plans with Unit members at regular Work-in-Progress sessions, receiving direct feedback to improve their proposals. ECRs are supported to submit strong fellowship and grant applications by the Epidemiology Science Strategy Group.

The Unit has a positive research environment that fosters engaged, fulfilled and productive team science and scientists. Our key areas of focus to achieve a vision of an inclusive, people-centred research culture are inclusive leadership, inclusive training and research career opportunities and inclusive research participation and involvement. The values that underpin the delivery of our strategy include: shared responsibility, data-driven improvement, building on best practice and continual learning. Specifically, we implement institutional initiatives throughout the professional journey of all staff and students. We use a diverse recruitment framework that includes a positive action statement encouraging under-represented groups to apply and mandate that all panel members have completed training in equality and diversity and unconscious bias. We encourage all staff to complete information on protected characteristics to monitor our EDI progress. The University of Cambridge has held a silver award since 2018 and joined Advance HE's Race Equality Charter in 2016 and received a bronze award in 2019.

Alongside institutional initiatives, we foster a culture that EDI is everyone’s business supported by Equality Champions from all staff groups who raise awareness of EDI issues in creative ways. We have a zero tolerance attitude to bullying and harassment and have implemented robust report and support mechanisms together with active bystander training to sustain a positive and equitable research culture.

We embed flexible practices to enable career support and progression. We promote access to a number of transparent University promotion and reward schemes relevant to different staff groups. We ensure fairness in authorship and promote a culture of team science throughout the working of the department. We align with DORA principles and have a strong commitment to open science across publications and data. We have a track record of involving the public in the research process through advisory panels recruited from diverse groups.

9. Detailed Budget

The finance section should provide a breakdown of costs associated with undertaking the research as described in the proposal.

1

Staff type	Lead Applicant

Name	Thomas Rance

Employing organisation	University of Cambridge

Grade	ST6

Starting point on scale	Not Applicable.

Increment Date	

Justification of Costs
ST6 Registrar annual income as per NHS: £63,152 + London weighting (20% with maximum of: £7736). National insurance calculated. Superannuation as per NHSBA pension. Pay scales for resident doctors in England (bma.org.uk)

Details of Posts and Salaries Costs

	Total (£)
Salary (£)	63152
Geographical Weighting (£)	7736
Other allowances (£)	
Superannuation (£)	1312.74
Nat. Insurance (£)	8722.16
Current Annual Costs (£)	**80922.9**

Staff Type: (Lead Applicant)-Dr Thomas Rance
Type of cost: HEI
% Full-time on this research: 100.0
Total months on this research: 36.0
Justification of Cost: Pay scales for resident doctors in England (bma.org.uk) National Insurance contributions calculated for 2025 tax year. Corroborated by University of Cambridge. London weighting applied as per current salary. Apprenticeship levy additional to this. ST6 Registrar annual income as per NHS: £63,152 + National insurance calculated. Superannuation as per NHSBA pension and Uni of Cam.

	Year 1	Year 2	Year 3	Total Cost (£)
Basic Salary	70425	70425	70425	211275
National Insurance	9551	9551	9551	28653
Superannuation	10212	10212	10212	30636
Geographical weighting / Other allowances (if applicable)				0

	Year 1	Year 2	Year 3	Total Cost (£)
Total HEI	72151	72151	72151	216453
Total NHS	0	0	0	0
Total Commercial	0	0	0	0
Total Other	0	0	0	0

Travel, Subsistence and Conference Fees

Type of cost: HEI
Description: Conference Fees
Conference: True
Item Description: RCPCH Conference
Justification of Cost: RCPCH Conference - 3 days - 420 Travel - estimate 100 train fare return - Conference typically held in Glasgow Accomodation - estimate 100 per night 3 days subsistence

	Year 1	Year 2	Year 3	Total Cost (£)
	1000	1000	1000	3000

Type of cost: HEI
Description: Journey Costs
Conference: False
Item Description: Travel
Justification of Cost: Travel for meetings and engagement with mentor/policy makers - Cambridge to London - twice per month. Estimate 44 pounds train fare.

	Year 1	Year 2	Year 3	Total Cost (£)
	1056	1056	1056	3168

Type of cost: HEI
Description: Subsistence
Conference: False
Item Description: Overseas visit
Justification of Cost: 6 week visit to British Columbia - Vancouver. Prices as per University of Cambridge guidance.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	6300	0	6300

Type of cost: HEI
Description: Journey Costs
Conference: False
Item Description: Overseas visit - travel
Justification of Cost: Return travel to Vancouver.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	1200	0	1200

	Year 1	Year 2	Year 3	Total Cost (£)
Travel, Subsistence and Conference Fees Total	2056	9556	2056	13668

Equipment

Type of cost: HEI
VAT relief to be applied: False
Description: Laptop
Price excluding VAT £: 1000
Expected life time: 36
Justification of Cost: My current laptop is over 10 years old. Given the work that this new role will require a new laptop with updated hardware and software with an increased processing ability is vital.

	Year 1 (exc VAT)	Year 2 (exc VAT)	Year 3 (exc VAT)	Total Cost (£) (exc VAT)
	1000	0	0	1000

	Year 1	Year 2	Year 3	Total Cost (£)
Equipment Total	1000	0	0	1000

Summary

	Travel (excl Conferences)	Travel (Conferences)	Equipment
Total HEI	8535	3000	800
Total NHS	0	0	0
Total Commercial	0	0	0
Total Other	0	0	0

Consumables

Description: Consumables for workstreams and dissemination
Type of cost: HEI
Please describe fully: Consumables for workstreams and dissemination
Justification of Cost: Consumables costed for in conjunction with UoC departmental advice.

	Year 1	Year 2	Year 3	Total Cost (£)
	1500	1500	1500	4500

	Year 1	Year 2	Year 3	Total Cost (£)
Consumables Total	1500	1500	1500	4500

Patient and Public Involvement

Description: PPIE development meetings
Type of cost: HEI
Please describe fully: PPIE development meetings
Justification of Cost: Development meeting: in-depth requirements & activity planning WS1 - Meeting to introduce population level survey project - Year 1 WS2 - Meeting to introduce hospital level collection project - Year 1 WS3 - Meeting to introduce major trauma data project - Year 2 Dissemination - Meeting to introduce ideas re dissemination of outputs - Year 3 Costing as per NIHR and CA:RING standard operating procedures. Run 2 workshops with bespoke participant recruitment (N=12), transcription/analysis and report for co-development of

WS1&2. Workshop facilitation: 4 x 2 hrs WS1: Co-development of survey WS2: Co-development of proforma PPIE with Young People's Advisory Group (N=20) to do a PPI activity for an hour, with own slides and activities already prepared WS3: Workshop to understand priorities for national trauma data and refine study Dissemination: Co-development of strategy for dissemination

	Year 1	Year 2	Year 3	Total Cost (£)
	200	100	100	400

Description: Workshop planning & materials preparation (16hrs)
Type of cost: HEI
Please describe fully: Workshop planning & materials preparation (16hrs)
Justification of Cost: Costing as per NIHR and CA:RING standard operating procedures. Run 2 workshops with bespoke participant recruitment (N=12), transcription/analysis and report for co-development of WS1&2. Workshop facilitation: 4 x 2 hrs WS1: Co-development of survey WS2: Co-development of proforma PPIE with Young People's Advisory Group (N=20) to do a PPI activity for an hour, with own slides and activities already prepared WS3: Workshop to understand priorities for national trauma data and refine study Dissemination: Co-development of strategy for dissemination

	Year 1	Year 2	Year 3	Total Cost (£)
	3200	0	0	3200

Description: Communication between members
Type of cost: HEI
Please describe fully: Communication between members
Justification of Cost: Communications with CA:RING members WS1- Review of study design and overview. Feedback. WS2 - Review of study design and overview. Feedback. Communications between CA:RING and participants in preparation of workshops. Compensation for time.

	Year 1	Year 2	Year 3	Total Cost (£)
	800	800	0	1600

Description: Final review meeting
Type of cost: HEI
Please describe fully: Final review meeting
Justification of Cost: Final review meeting ahead of each workshop (1.5hrs x 2) Ensure preparation for workshops to codesign WS1& WS2.

	Year 1	Year 2	Year 3	Total Cost (£)
	150	150	0	300

Description: Final adjustments following review meeting
Type of cost: HEI
Please describe fully: Final adjustments following review meeting
Justification of Cost: Costing as per NIHR and CA:RING standard operating procedures. Run 2 workshops with bespoke participant recruitment (N=12), transcription/analysis and report for co-development of WS1&2. Workshop facilitation: 4 x 2 hrs WS1: Co-development of survey WS2: Co-development of proforma PPIE with Young People's Advisory Group (N=20) to do a PPI activity for an hour, with own slides and activities already prepared WS3: Workshop to understand priorities for national trauma data and refine study Dissemination: Co-development of strategy for dissemination

	Year 1	Year 2	Year 3	Total Cost (£)
	100	100	0	200

Description: Workshop facilitation
Type of cost: HEI
Please describe fully: Workshop facilitation
Justification of Cost: Costing as per NIHR and CA:RING standard operating procedures. Run 2 workshops with bespoke participant recruitment (N=12), transcription/analysis and report for co-development of WS1&2. Workshop facilitation: 4 x 2 hrs WS1: Co-development of survey WS2: Co-development of proforma PPIE with Young People's Advisory Group (N=20) to do a PPI activity for an hour, with own slides and activities already prepared WS3: Workshop to understand priorities for national trauma data and refine study Dissemination: Co-development of strategy for dissemination

	Year 1	Year 2	Year 3	Total Cost (£)
	200	400	200	800

Description: Post-workshop debrief with researcher
Type of cost: HEI
Please describe fully: Post-workshop debrief with researcher
Justification of Cost: Run 2 workshops with bespoke participant recruitment (N=12), transcription/analysis and report for co-development of WS1&2. Workshop facilitation: 4 x 2 hrs WS1: Co-development of survey WS2: Co-development of proforma PPIE with Young People's Advisory Group (N=20) to do a PPI activity for an hour, with own slides and activities already prepared WS3: Workshop to understand priorities for national trauma data and refine study Dissemination: Co-development of strategy for dissemination

	Year 1	Year 2	Year 3	Total Cost (£)
	50	100	50	200

Description: Workshop transcription
Type of cost: HEI
Please describe fully: Workshop transcription
Justification of Cost: Costing as per NIHR and CA:RING standard operating procedures. Run 2 workshops with bespoke participant recruitment (N=12), transcription/analysis and report for co-development of WS1&2. Workshop facilitation: 4 x 2 hrs WS1: Co-development of survey WS2: Co-development of proforma PPIE with Young People's Advisory Group (N=20) to do a PPI activity for an hour, with own slides and activities already prepared WS3: Workshop to understand priorities for national trauma data and refine study Dissemination: Co-development of strategy for dissemination

	Year 1	Year 2	Year 3	Total Cost (£)
	1200	2400	1200	4800

Description: Workshop analysis & reporting
Type of cost: HEI
Please describe fully: Workshop analysis & reporting
Justification of Cost: Costing as per NIHR and CA:RING standard operating procedures. Run 2 workshops with bespoke participant recruitment (N=12), transcription/analysis and report for co-development of WS1&2. Workshop facilitation: 4 x 2 hrs WS1: Co-development of survey WS2: Co-development of proforma PPIE with Young People's Advisory Group (N=20) to do a PPI activity for an hour, with own slides and activities already prepared WS3: Workshop to understand priorities for national trauma data and refine study Dissemination: Co-development of strategy for dissemination

	Year 1	Year 2	Year 3	Total Cost (£)
	4000	4000	0	8000

Description: Processing participant payments
Type of cost: HEI
Please describe fully: Processing participant payments
Justification of Cost: Processing participant payments (2hrs x 4)

	Year 1	Year 2	Year 3	Total Cost (£)
	200	400	200	800

Description: Participant payments
Type of cost: HEI
Please describe fully: Participant payments
Justification of Cost: Participant payments (£25 x 2hrs x 12 people x 4 workshops) Costing as per NIHR and CA:RING standard operating procedures. Run 2 workshops with bespoke participant recruitment (N=12), transcription/analysis and report for co-development of WS1&2. Workshop facilitation: 4 x 2 hrs WS1: Co-development of survey WS2: Co-development of proforma PPIE with Young People's Advisory Group (N=20) to do a PPI activity for an hour, with own slides and activities already prepared WS3: Workshop to understand priorities for national trauma data and refine study Dissemination: Co-development of strategy for dissemination

	Year 1	Year 2	Year 3	Total Cost (£)
	600	1200	600	2400

Description: Admin time
Type of cost: HEI
Please describe fully: Admin time
Justification of Cost: Admin time x 4 Participant payments (£25 x 2hrs x 12 people x 4 workshops) Costing as per NIHR and CA:RING standard operating procedures. Run 2 workshops with bespoke participant recruitment (N=12), transcription/analysis and report for co-development of WS1&2. Workshop facilitation: 4 x 2 hrs WS1: Co-development of survey WS2: Co-development of proforma PPIE with Young People's Advisory Group (N=20) to do a PPI activity for an hour, with own slides and activities already prepared WS3: Workshop to understand priorities for national trauma data and refine study Dissemination: Co-development of strategy for dissemination

	Year 1	Year 2	Year 3	Total Cost (£)
	200	400	200	800

	Year 1	Year 2	Year 3	Total Cost (£)
Patient and Public Involvement Total	10900	10050	2550	23500

Other Direct Costs

Description: WS1 - Population level survey
Type of cost: HEI
Please describe fully: WS1 - Population level survey
Justification of Cost: Facilitated by YouGov to ensure representative and inclusive data with EDI incorporated. Under expertise from previous iterations of play surveys. Indicative costs from previous projects but exact costing will be done when project finalised.

	Year 1	Year 2	Year 3	Total Cost (£)
	40000	0	0	40000

Description: Research Costs
Type of cost: HEI
Please describe fully: Research Costs
Justification of Cost: As per Uni of Cam guidance from SoECAT.

	Year 1	Year 2	Year 3	Total Cost (£)
	4399	0	0000000000000000	4399

	Year 1	Year 2	Year 3	Total Cost (£)
Other Direct Costs Total	44399	0	0	44399

Training And Development

Type of cost: HEI
Description: Short Courses
Item Description: Introductory Course in Epidemiology & Medical Statistics and Advanced Course in Epidemiological Analysis - LSHTM
Justification of Cost: Vital training for expertise in epidemiology Introductory course in year 1 followed by advance course in year 2.

	Year 1	Year 2	Year 3	Total Cost (£)
	3537	2358	0	5895

Type of cost: HEI
Description: Applicant PhD Tuition Fees
Item Description: PhD Fees
Justification of Cost: PhD Fees as per University website: £ 10,356 per year https://www.postgraduate.study.cam.ac.uk/finance/fees

	Year 1	Year 2	Year 3	Total Cost (£)

	10356	10356	10356	31068

Type of cost: HEI
Description:
Item Description: Data management training
Justification of Cost: as per advice from primary PhD supervisor - data management course at University of Oxford: https://www.conted.ox.ac.uk/courses/an-overview-of-data-science?code=O22P759COW

	Year 1	Year 2	Year 3	Total Cost (£)
	260	0	0	260

Type of cost: HEI
Description: Short Courses
Item Description: Big data epidemiology
Justification of Cost: Big data epidemiology at the University of Oxford. This will help with processing large amount of data in WS3 for the retrospective NMTR data. https://www.conted.ox.ac.uk/courses/big-data-epidemiology?code=O24C043B9Y

	Year 1	Year 2	Year 3	Total Cost (£)
	0	2570	0	2570

	Year 1	Year 2	Year 3	Total Cost (£)
Training And Development Total	14153	15284	10356	39793

Total Direct Costs

	Year 1	Year 2	Year 3	Total Cost (£)
Total Direct Costs	134388	104520	87592	326500

Indirect costs

HEI Indirect costs

Type of cost: Estates Charges
Name of Higher Education Institute: University of Cambridge
Justification of Cost: As per university financial guidance

	Year 1	Year 2	Year 3	Total Cost (£)
	20304	20304	20304	60912

Type of cost: Other Indirect Costs
Name of Higher Education Institute: University of Cambridge
Justification of Cost: Infrastructure Technician Costs

	Year 1	Year 2	Year 3	Total Cost (£)
	1786	1786	1786	5358

Type of cost: Other Indirect Costs
Name of Higher Education Institute: University of Cambridge
Justification of Cost: University of Cambirdge Indirect Costs - as per University guidance

	Year 1	Year 2	Year 3	Total Cost (£)
	74984	74984	74984	224952

	Year 1	Year 2	Year 3	Total Cost (£)
HEI Indirect costs Total	97074	97074	97074	291222

Commercial Indirect Costs
No costs have been added

Other Partner Organisation Indirect costs
No costs have been added

	Year 1	Year 2	Year 3	Total Cost (£)
Total Indirect Costs	77659	77659	77659	232977

Research Costs Requested From Funder

	Direct costs	Indirect costs/Overheads	Amount requested
Total Higher Education Institution Costs (80%)	283707	232977	516684
Total Higher Education Institution Costs (100%)s	42793	0	42793
Total Commercial Costs	0	0	0
Total Other Parther Organisation Costs	0	0	0
Total NHS Research Costs	0	0	0
Total Research Costs Requested From Funder	326500	232977	559477

Summary

	Year 1	Year 2	Year 3	Total Cost (£)
Salaries	**72151**	**72151**	**72151**	**216453**
Travel-subsistence-and-conference-fees	**1845**	**7845**	**1845**	**11535**
Equipment	**800**	**0**	**0**	**800**
Consumables	**1200**	**1200**	**1200**	**3600**
Patient-and-Public-Involvement	**8720**	**8040**	**2040**	**18800**
Other-Direct-Costs	**35519**	**0**	**0**	**35519**
Training And Development	**14153**	**15284**	**10356**	**39793**
HEI-Indirect-costs	**77659**	**77659**	**77659**	**232977**
Commercial-Indirect-Costs	**0**	**0**	**0**	**0**
Other-Partner-Organisation-Indirect-costs	**0**	**0**	**0**	**0**
Total	**212047**	**182179**	**165251**	**559477**

NHS Support Costs
No costs have been added

Have you discussed and agreed these costs with the Lead Network?	Yes

NHS Treatment Costs
No costs have been added

Have you discussed and agreed the NHS costs with the Lead Trust?	Yes

Is the patient/service user care being provided different from the usual standard treatment for the condition?
Yes

Usual Treatment Costs
No costs have been added

Excess Treatment costs/(savings)

	Cost per patient(£)	Year 1	Year 2	Year 3	Total Cost (£)
Total NHS Excess Treatment costs/(savings)	0	0	0	0	0

	Total (£)
NHS Support Costs requested from networks	0
NHS Excess Treatment costs/savings	0
TOTAL NHS SUPPORT AND TREATMENT COSTS	0

Total Funding Required (including non-research costs)

	Total (£)
Total Research Costs Requested from Funder	559477
Total NHS (Support and Excess Treatment) Costs	0
Total Funding Required	559477

10. Uploads

Please upload the following documents, if necessary.

When uploading a document to your application, please ensure that it is in an appropriate file format. It is not possible to upload JPEG images, and with the exception of the SoECAT Form, which should be uploaded as a .csv file or Excel spreadsheet, all files should be uploaded in Adobe PDF or Word DOC/DOCX format. Please refer to the Guidance Notes for further information on the attachments required.

(References - NIHR R13 References.pdf) is included as an appendix within this file.

(Research Timetable - PhD Gantt chart.pdf) is included as an appendix within this file.

(SoECAT Template - SOECAT - Funder Export 12.24.xlsx) is included as an appendix within this file.

11. Participants and Signatory

Participants
New participants must confirm their participation on your application before you will be able to select the 'SUBMIT' option. Please see the Submission Process Flow Diagram and the Applicant Guidance Notes for further information.

You are required to supply the names and email addresses (if not already registered with this online application system) of each required participant. Everyone **newly** named in this section will be acting as a 'participant' to your application and will need to agree to be part of this application.

By confirming participation, participants are acknowledging their involvement and input into this application, and agreeing to be involved in the proposed fellowship. You must ensure all **new** participants are happy for your application to be submitted before submitting it on the online system.

It is permissible for a single individual to serve as a signatory and a participant, but they will need to be added separately under each heading.

Details of the individual who will be required to approve your application after submission should be entered in the 'Participants and Signatory' section. Please note that it is acceptable for a Head of Department to also act as a research support role. If this is the case, they will need to be included in this section for both roles.

Participants must confirm their participation on your application before you will be able to press the submit button. Please see the 'Application Submission Process Flow Diagram' in the guidance notes for further information.

Doctoral Primary Supervisor

Doctoral Primary Supervisor Participant

Title	Dr
First Name	Esther
Last Name	van Sluijs
Position	MRC Programme Leader

Supervisor

Supervisor Participant

Title	Dr
First Name	Kathryn
Last Name	Hesketh
Position	Research Associate

Supervisor Participant

Title	Professor
First Name	Paul
Last Name	Ramchandani

Position	Professor

Administrative Authority or Finance Officer

Administrative Authority or FO Participant

Title	Ms
First Name	Kiyal
Last Name	Eresen
Position	

Signatory
Participants must complete their actions on your application before submission; the signatory must also approve your application after you select the 'submit' option but BEFORE the application submission deadline.

You are required to supply the name and email address (if not already registered on the ARAMIS application system) of the individuals who will be 'signing off' your application. Once their contact details have been entered below, these individuals will be invited to log in and confirm their participation in your application.

The signatory must ALSO approve the application AFTER you have selected the 'SUBMIT' option but BEFORE the application submission deadline. Please see the Submission Process Flow Diagram and Applicant Guidance Notes for further information.

The signatory approval will result in the application being fully submitted to the NIHR. All parties (applicant, participants and signatory) will be notified of this via an automated system-generated email.
NIHR will not accept any applications unless fully approved by your signatory before the 1:00 pm deadline on Thursday 16th January 2025.

Host Organisation - Heads of Department

Head of Department or Senior Manager

Title	Professor
First Name	Nick
Last Name	Wareham
Position	Director

12. Acknowledgement, Review and Submit

Conflict checks: Please declare any conflicts or potential conflicts of interest that you may have in undertaking this research, including any relevant, non-personal & commercial interest that could be perceived as a conflict of interest.
There are no known conflicts of interest.

By completing this form, the programme reserves the right to share, in confidence, details of your application with other parts of the NIHR and with the Department of Health and Social Care.

Agreement to Terms and Conditions: I confirm that I have secured all necessary licences and ethical approvals in relation to the research and will abide by the terms of those licences and approvals in carrying out the research. I confirm that I have read the NIHR Carbon Reduction Guidelines and, where possible, have taken steps to reduce the carbon emissions generated by this research. I confirm that all material within this application, with the exception of sections that other participants are required to complete, is my own work and I have taken steps to ensure that no parts of the application have been inappropriately plagiarised. I undertake to conduct the research according to such guidelines for good research practice as the Host Organisation may from time to time lay down according to the NIHR Research Governance Framework. If you are applying for an NIHR award which comes with partnership funding from another organisation (e.g. a charity or industry partner) your information (including your NIHR Application Form and subsequent progress/annual reports) may be shared with the partner organisation as part of the review process and ongoing monitoring of the award. In completing this, I declare that, to the best of my knowledge, the information provided in this application, is true, accurate and complete. I have read the Guidance Notes and agree to accept the process by which an application is assessed and agree to abide by the conditions under which an award may be granted.
☒Confirmed

Appendices

1. NIHR R13 References.pdf
2. PhD Gantt chart.pdf
3. SOECAT - Funder Export 12.24.xlsx

Appendix #1: NIHR R13 References.pdf

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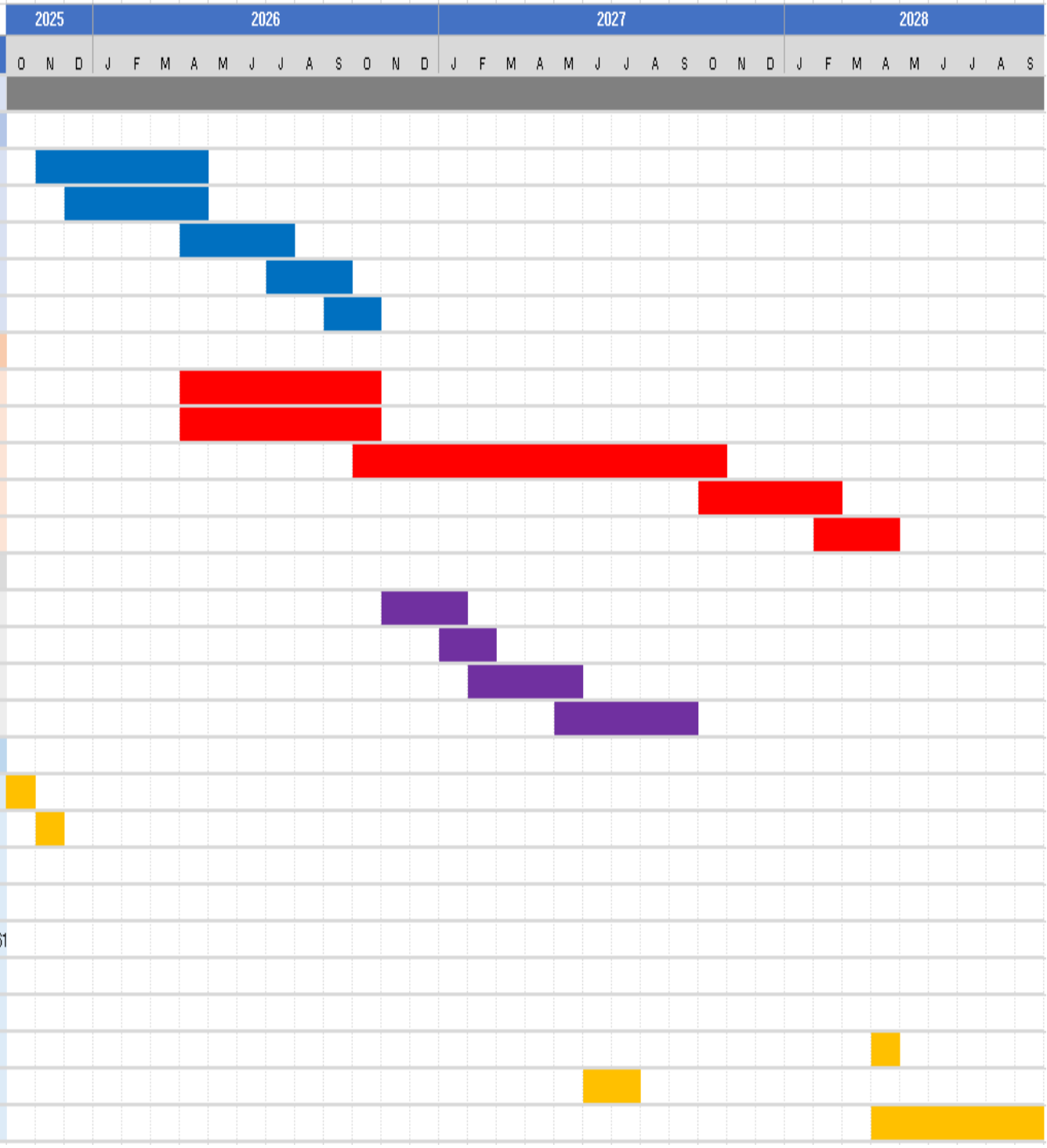
Appendix #2: PhD Gantt chart.pdf

Exploring the asssociation between Adventurous Play (AP) and injuries in children aged 0-11 in the UK

University of Cambridge

Dr Tom Rance

TASK DESCRIPTION	PLAN START	PLAN END
PhD Start date	01/10/2025	01/09/2028
Work Stream 1		
PPIE	1/11/25	1/4/26
Survey development	1/12/25	1/4/26
Data collection	1/4/26	1/7/26
Data analysis	1/7/26	1/9/26
Write-up & Dissemination	1/9/26	1/10/26
Work Stream 2		
PPIE	1/4/26	1/10/26
Data collection plan and pro forma development	1/4/26	1/10/26
Data Collection	1/10/26	1/10/27
Data analysis	1/10/27	1/2/28
Write-up & Dissemination	1/2/28	1/4/28
Work Stream 3		
PPIE	1/11/26	1/1/27
Data access and analysis plan	1/1/27	1/2/27
Data analysis	1/2/27	1/5/27
Write-up & Dissemination	2/5/27	1/9/27
Training & Development		
PPIE Training	20/10/25	23/10/25
EDI Training	3/11/25	6/11/25
Introductory Epidemiology Course - LSHTM	dates to be released	Aim to complete prior to 04/2026
Advanced Epidemiology Course - LSHTM	dates to be released	Aim to complete prior to 04/2027
Data management and statistical analysis - Oxford	dates to be released	Aim to complete prior to data analysis for WS1
Big data epidemiology - Oxford	dates to be released	Aim to complete for data analysis for WS3
PGCert in Teaching and Learning in Higher Education	5 days flexible + coursework	flexible timing
Leadership for Population Health	1/4/28	14/4/28
Research Visit - UBC	1/6/27	14/7/27
Thesis Writing	1/4/28	1/9/28



Appendix #3: SOECAT - Funder Export 12.24.xlsx